# Results: Citation Network

## RQ4: Citation Geometry I

Resolving each record's references against the corpus yields an intra-corpus citation graph (built and analyzed with NetworkX (Hagberg et al. 2008)) of 2234 nodes and 8,623 edges across 1409 connected components, with a graph density of 0.17% and a mean in-degree of 3.9. Of 38,489 total outgoing references, 22.4% resolve to another record inside the corpus — a resolution rate that reflects how self-contained the retrieved slice of the literature is rather than the underlying citation density of any single work.

## RQ4: Citation Geometry I

The citation network has 1416 communities (detected by modularity optimization), a maximum in-degree of 163 (the most-cited paper within the corpus), and a maximum out-degree of 143 (the paper that cites the most other corpus members).

## Centrality Analysis I

Centrality scores (PageRank (Page et al. 1999) and HITS) and modularity-based community detection (Clauset et al. 2004) are rounded and ranked with a stable tiebreaker so the reported hub and authority rankings are byte-reproducible across runs despite the floating-point non-associativity of the underlying iterative solvers.

**Table 4. Top 5 papers by PageRank.**

# Centrality Analysis I

| Rank | DOI | PageRank |
|---|---|---|
| 1 | 10.1177/026988110001400107 | 0.036943 |
| 2 | 10.1212/wnl.54.5.1166 | 0.026949 |
| 3 | 10.1523/jneurosci.21-05-01787.2001 | 0.013668 |
| 4 | 10.1523/jneurosci.20-22-08620.2000 | 0.011349 |
| 5 | 10.4088/jcp.v61n0510 | 0.008136 |

**Table 5. Top 5 authority papers (HITS).**

# Centrality Analysis I

| Rank | DOI | Authority |
|---|---|---|
| 1 | 10.1038/sj.npp.1301534 | 0.017700 |
| 2 | 10.1007/s00213-002-1250-8 | 0.016311 |
| 3 | 10.1124/jpet.106.106583 | 0.015144 |
| 4 | 10.1523/jneurosci.21-05-01787.2001 | 0.014734 |
| 5 | 10.1001/jama.2009.351 | 0.014481 |

**Table 6. Top 5 hub papers (HITS).**

# Centrality Analysis I

| Rank | DOI | Hub |
|---|---|---|
| 1 | 10.3389/fnins.2021.656475 | 0.011968 |
| 2 | 10.1016/bs.apha.2023.10.006 | 0.011596 |
| 3 | 10.1038/sj.npp.1301534 | 0.010886 |
| 4 | 10.1080/08897077.2019.1700584 | 0.010562 |
| 5 | 10.1007/s00213-013-3232-4 | 0.009896 |

## Centrality Analysis I

The most influential paper by PageRank (DOI 10.1177/026988110001400107) is a foundational work that anchors the citation structure — its high authority score confirms it is frequently cited by other corpus members. Hub papers, which cite many other corpus members, serve as integrative reviews or meta-analyses that connect disparate threads of the literature.

# Centrality Analysis I

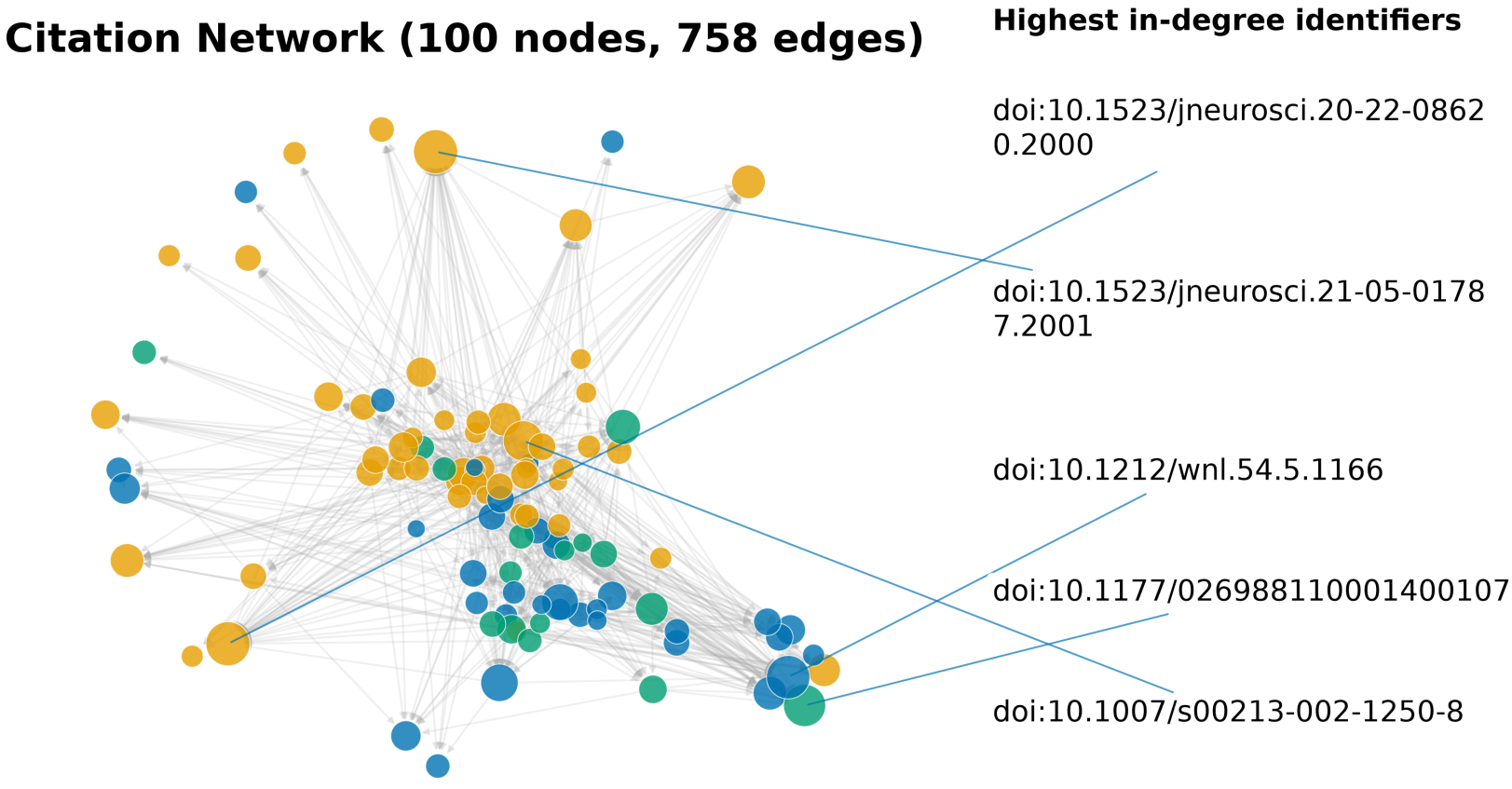


Figure 17: Citation network for Modafinil. Nodes represent papers; directed edges represent citation links. Node colours indicate community membership (1416 communities detected by modularity optimization). Layout uses a spring-based algorithm with a fixed seed for reproducibility.

# Centrality Analysis I

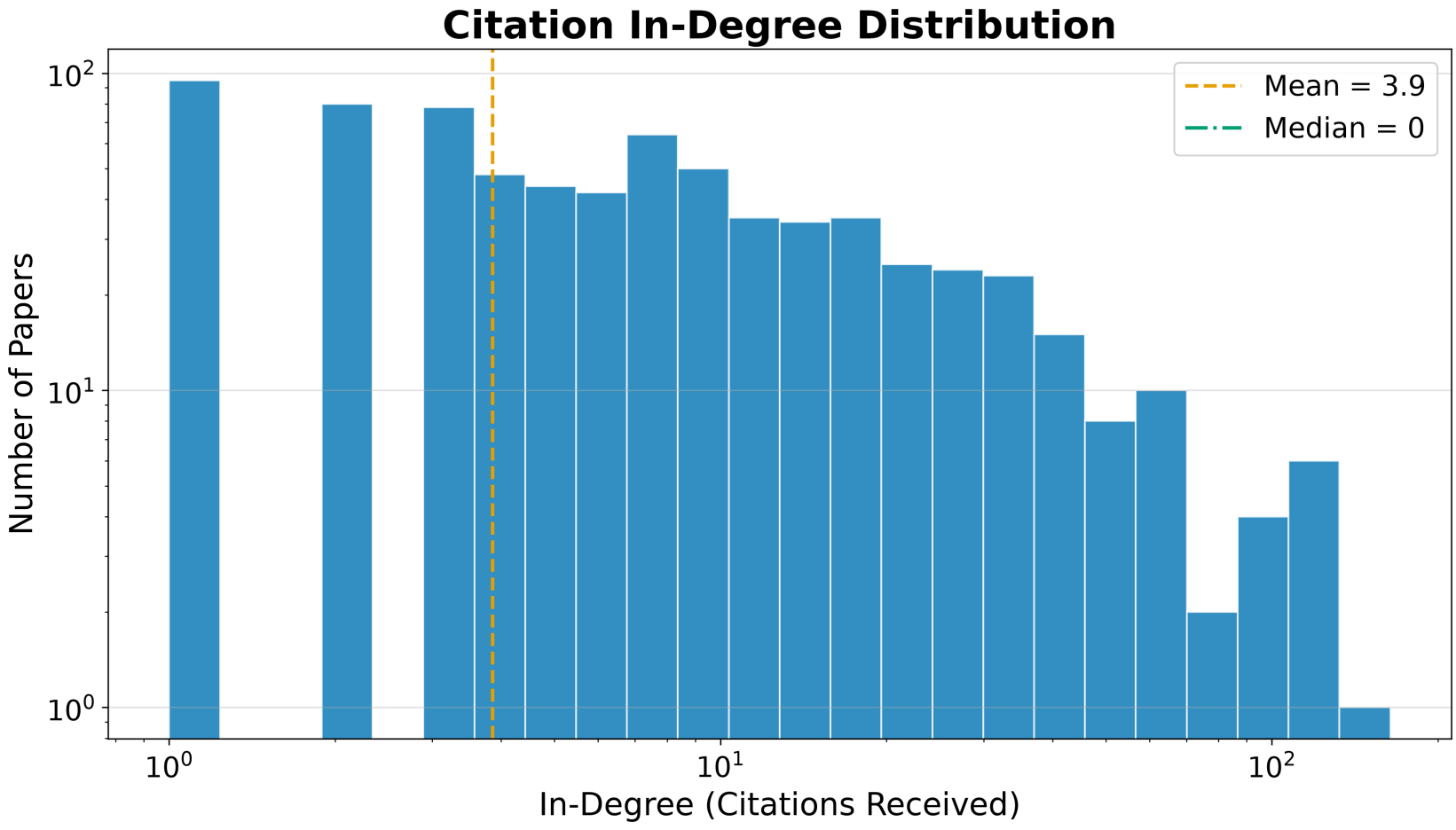


Figure 18: Degree distribution for the Modafinil citation network. The histogram shows the frequency of each in-degree value on a log-linear scale, revealing the heavy-tailed structure characteristic of citation networks.

## Centrality Analysis I

The heavy-tailed degree distribution is characteristic of citation networks: a small number of highly-cited papers anchor the structure, while the long tail of low-degree nodes represents newer or peripheral works. The low graph density (0.17%) reflects the sparsity of intra-corpus citation links — most papers cite works outside the retrieved slice, which is expected for a max-results-capped retrieval.

## Advanced Network Metrics I

Beyond PageRank and HITS, the network analysis computes betweenness centrality (which papers bridge different communities), closeness centrality (which papers are near all others), degree assortativity (do highly-cited papers cite other highly-cited papers?), and average clustering coefficient (how tightly knit are local neighborhoods).

The degree assortativity coefficient is -0.0598, and the average clustering coefficient is 0.1029. A negative assortativity indicates that highly-cited papers tend to cite less-cited papers (dissortative mixing), which is typical of citation networks where review papers (high in-degree) cite many primary studies (low in-degree).

**Table 7. Top 5 papers by betweenness centrality.**

# Advanced Network Metrics I

| Rank | DOI | Betweenness |
|---|---|---|
| 1 | 10.1038/sj.npp.1301534 | 0.005234 |
| 2 | 10.4088/jcp.v67n0406 | 0.003451 |
| 3 | 10.1016/j.neuropharm.2012.07.011 | 0.002530 |
| 4 | 10.1002/14651858.cd006788.pub3 | 0.002289 |
| 5 | 10.2165/00003495-200868130-00003 | 0.001894 |

Papers with high betweenness centrality serve as bridges between different topical communities in the citation network — their removal would fragment the graph into disconnected components. These bridging papers are often review articles or methodological papers that connect disparate research threads.